Heriot-Watt University

Department of Actuarial Mathematics and Statistics

Life Insurance Mathematics B

Tutorial 8

1. Calculate the following assuming the mortality of the AM92 table, and describe each expression in words:

   (a) ${}_{10}p_{30:40}$

   (b) $q_{30:40}$

   (c) $\mu_{40:50}$

   (d) ${}_{10}p_{[30]:[40]}$

   (e) $q_{[30]:[40]}$

   (f) $\mu_{[40]:[50]}$

   (g) $\mu_{[40]+1:[60]+1}$

   (h) ${}_{3}|q_{[30]+1:[40]+1}$

   Hint: Numerical answers are 0.97853, 0.001526, 0.003274, 0.97887, 0.001264, 0.002293, 0.008129, 0.001976

2. Let T_x and T_y be the independent random future lifetimes of two lives age x and y, and define $T_{\min} = \min[T_x, T_y]$ and $T_{\max} = \max[T_x, T_y]$.

   (a) Derive an expression for the density of $T_{\max}$.

   (b) Define $\mathring{e}_{xy} = \mathrm{E}[T_{\min}]$. Show that:

   $$\mathring{e}_{xy} = \int_0^\infty {}_tp_{xy}dt.$$

   (c) Show that $\mathrm{Cov}\left(T_{\min}, T_{\max}\right) = \left(\mathring{e}_x - \mathring{e}_{xy}\right)\left(\mathring{e}_y - \mathring{e}_{xy}\right)$

   (d) Further let K_{min} be the integer part of $T_{\min}$ and define $e_{xy} = \mathrm{E}[K_{\min}]$. Show that

      i. $e_{xy} = \sum_{t=1}^{\infty} {}_tp_{xy}$. and

      ii. $\mathring{e}_{xy} \approx e_{xy} + \frac{1}{2}$.

   (e) Derive an expression for the 'force of mortality' associated with T_{max}, denoted $\mu_{\overline{x:y}}(t)$. What is its value at $t = 0$? Explain this result.

3. Given that $l_{xy} = 10,000$, $l_{x+10:y} = 9,600$, and $l_{x:y+10} = 9,200$ calculate the probability that, of the two independent lives aged x and y, exactly one will survive for 10 years.

   Hint: The answer is 0.1136

4. Show that:

   (a) $\ddot{a}_{xy} = \sum_{k=0}^{\infty} v^k {}_kp_{xy}$.

   (b) $\ddot{a}_{xy:\overline{n}|} = \ddot{a}_{x:\overline{n}|} + \ddot{a}_{y:\overline{n}|} - \ddot{a}_{\overline{xy}:\overline{n}|}$.

   (c) $A_{\overline{xy}} = A_x + A_y - A_{xy}$.

5. Derive an expression for the variance of the random variable $v^{K_{max}+1}$.

6. For a male aged 70 exact and a female aged 67 exact, who are subject to the mortality of the PMA92 and PFA92 tables respectively, with interest of 4% per annum, find:

   (a) $\ddot{a}_{70:67}$
   (b) $\ddot{a}^{(12)}_{70:67}$
   (c) $\ddot{a}_{70:67:\overline{10}|}$
   (d) $\ddot{a}^{(12)}_{70:67:\overline{10}|}$
   (e) $\ddot{a}_{\overline{70:67}}$
   (f) $\ddot{a}^{(12)}_{\overline{70:67}}$

   Hint: The answers are 10.233, 9.775, 7.458, 7.204, 15.44, 14.982

7. State in words the meanings of the symbols A_{xy}, $A_{\overline{xy:n}|}$ and $\bar{A}_{xy:\overline{n}|}$. Prove that:

   (a) $A_{\overline{xy:n}|} = 1 - d\ddot{a}_{\overline{xy:n}|}$
   (b) $\bar{A}_{xy:\overline{n}|} = 1 - \delta\bar{a}_{xy:\overline{n}|}$.